Antenna AI Predictor

Revolutionising Antenna Design Through Machine Learning

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Transforming Traditional Design Methods

The Challenge

Traditional antenna design relies on trial-and-error processes where each geometric modification demands complex electromagnetic simulations. These simulations can consume minutes or even hours, creating significant bottlenecks in the design cycle.

The Solution

Our AI-driven approach replaces slow EM solvers with neural networks trained on physics-based data, enabling millisecond predictions whilst maintaining high accuracy.

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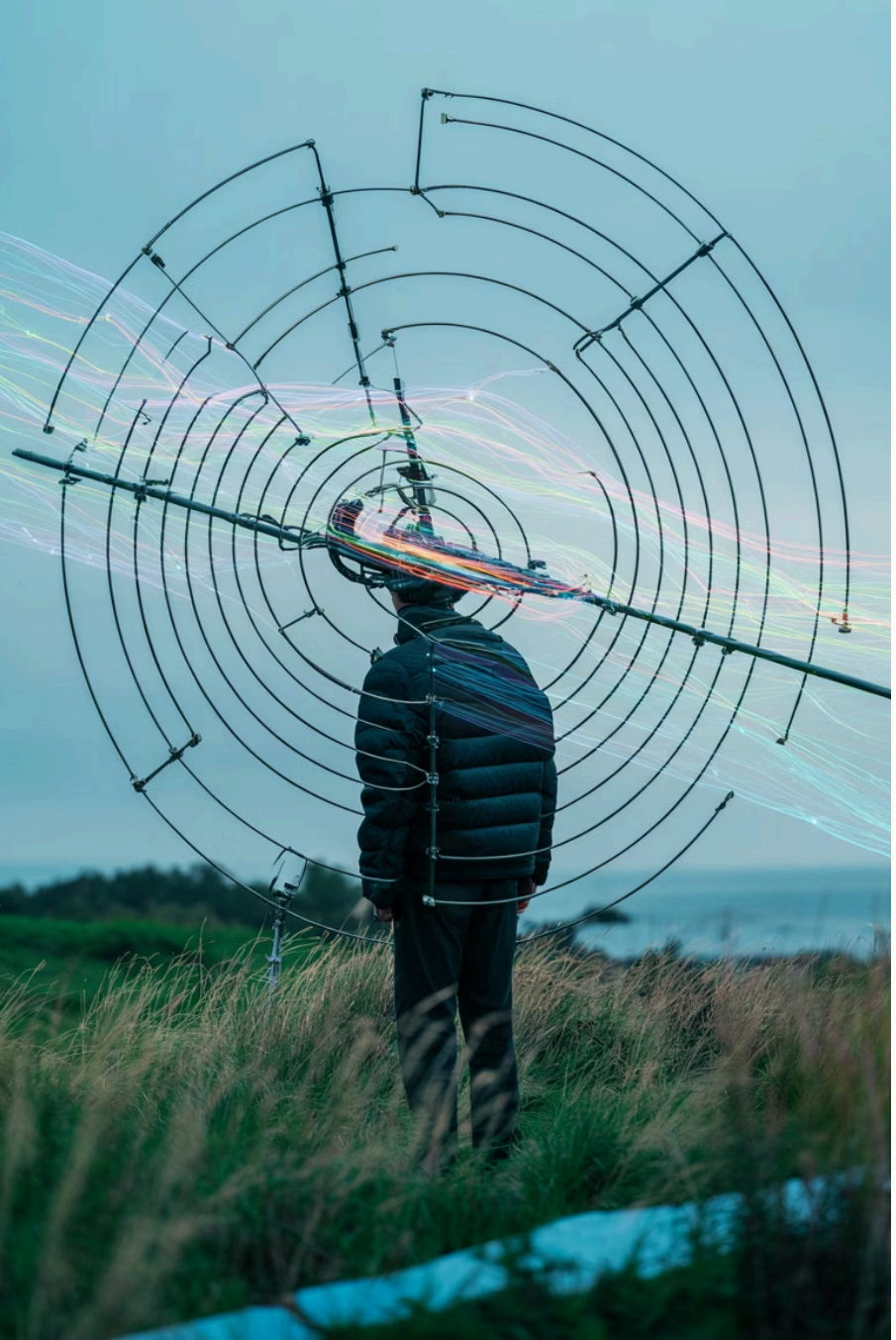

Project Overview

AI-Powered Design

Neural networks learn underlying physics from simulation data, creating a "virtual solver"

Massive Speedup

Over 1,000,000× faster than traditional electromagnetic solvers

High Accuracy

Typically achieves >95% accuracy compared to physics-based simulation

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Complete Workflow Process

Data Generation

Physics-based simulation using OpenEMS to create training dataset

ML Training

Neural network training with PyTorch framework

Verification

Accuracy and speed comparison testing

Visualization

Results analysis and performance plots

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Step 1: Data Generation

Physics-Based Simulation

Tools: OpenEMS (open-source FDTD solver) and CSXCAD for geometric modelling

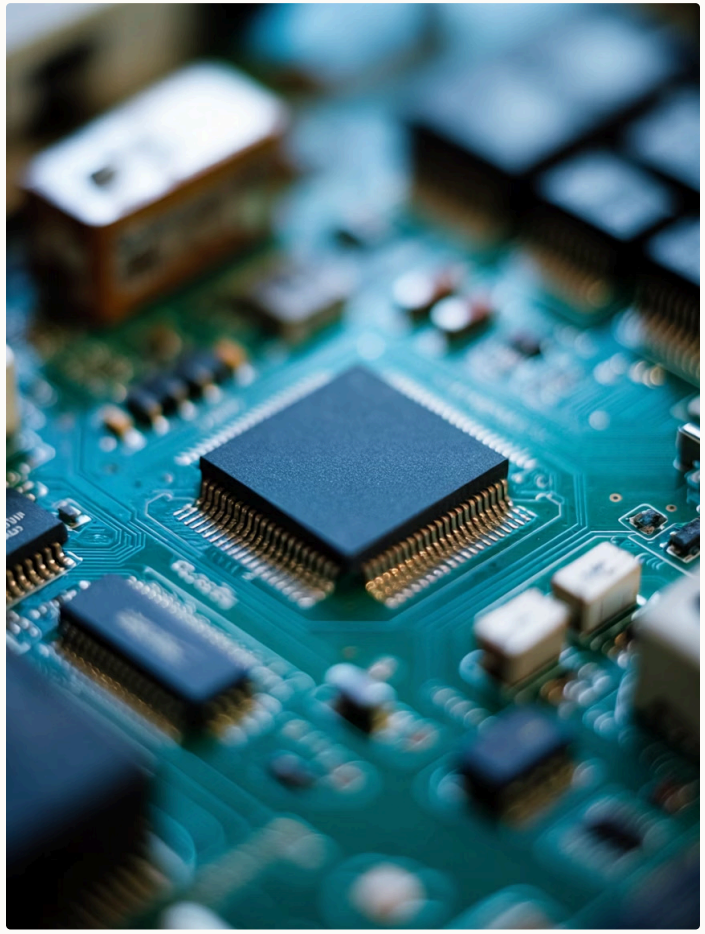
Antenna Configuration

Microstrip Patch Antenna on FR4 substrate with three variable parameters:

- Length (L)
- Width (W)
- Feed Position (FP)

Output Metrics

For each geometric variation, the solver calculates S11 (Return Loss) and Far-field Gain, producing bandwidth in GHz and maximum gain in dBi.



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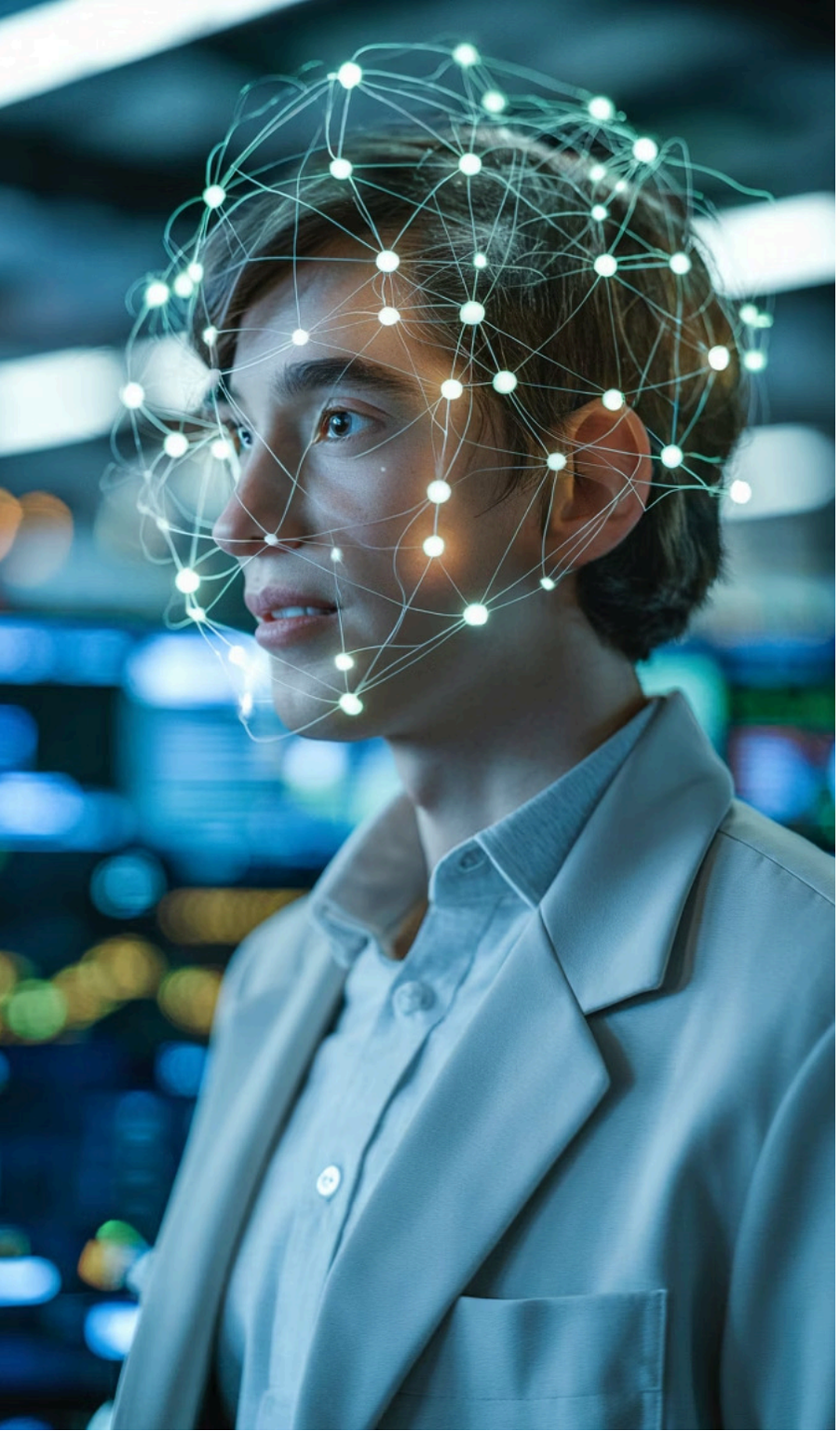

Step 2: Machine Learning Training

1

Forward Predictor

Takes physical dimensions (L, W, FP) as input and predicts performance metrics (Bandwidth, Gain). Functions as a "Virtual Solver" that replaces traditional simulation.

2

Inverse Encoder

Takes target performance goals (BW, Gain) as input and predicts required physical dimensions. Enables automated design from specifications.

3

Training Configuration

PyTorch framework with Adam Optimiser and Mean Squared Error (MSE) loss function, trained for 2,000 epochs to ensure high fidelity to physical data.

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Step 3: Verification & Comparison

Performance Testing

60s

Traditional Solver

OpenEMS simulation time

1µs

AI Prediction

Neural network inference time

>95%

Accuracy

Match to physics-based solver

Testing Protocol: AI receives target performance goal (e.g., 0.5 GHz bandwidth), predicts dimensions, and verifies against physics-based simulation to confirm design accuracy.

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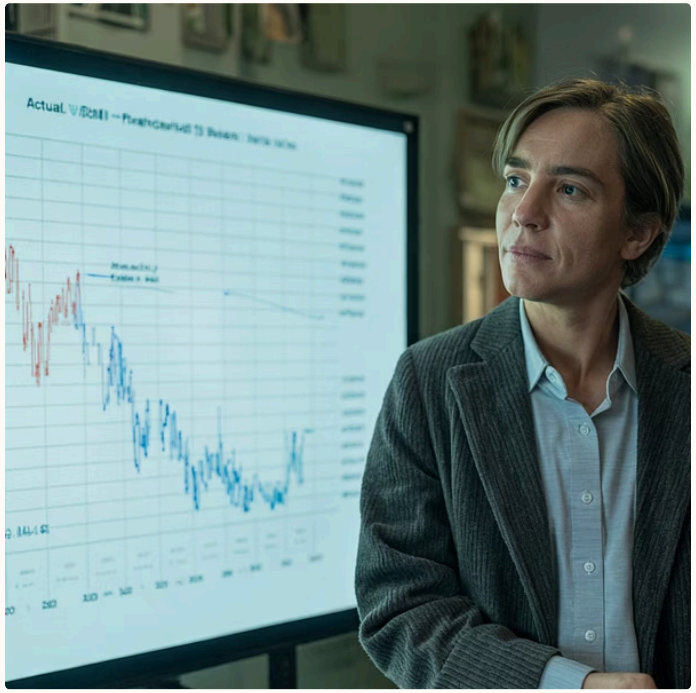
Step 4: Results Visualization



AI Accuracy Plots

Visualises actual versus predicted values for bandwidth and gain, demonstrating model fidelity

Speedup Chart

Logarithmic representation of efficiency gains, highlighting 1,000,000× improvement

S11 Resonance Plot

Generates return loss curve showing antenna resonance "dip" at target frequency

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File Architecture Summary

generate_dataset_openems.py

The Data Engine

Runs physical simulations using OpenEMS to generate comprehensive training dataset with geometric variations and corresponding performance metrics

train_predictor.py

The Brain

Implements neural network training pipeline, configuring and training both forward predictor and inverse encoder models

visualize_results.py

The Reporting Tool

Generates comprehensive accuracy plots, speed comparison charts, and resonance visualisations for validation

dataset_patch.csv

The Knowledge Base

Comma-separated values file containing all simulation results: physical dimensions, bandwidth measurements, and gain values

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Transforming Antenna Design

Millisecond Design Cycles

Replace hours of simulation with instantaneous predictions, accelerating iteration and optimisation

Accessible Expertise

Democratise antenna design by reducing reliance on specialised simulation expertise and computational resources

Scalable Solutions

Apply the same methodology to diverse antenna types and frequency ranges with minimal retraining

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